

MATHEMATICS**Maximum Marks: 80****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the paper.
2. You must NOT start writing during reading time.
3. The question paper has 9 printed pages.
4. It consists of twenty questions and four sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in three questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

Instruction to Supervising Examiner

Kindly read aloud the instructions given above to all the candidates present in the examination hall.

SECTION A – 20 MARKS**Question 1**

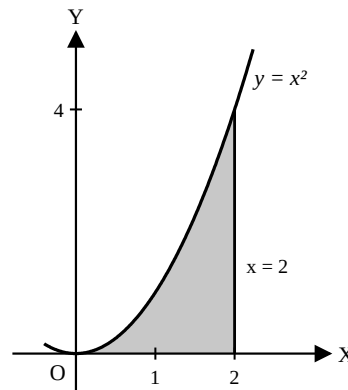
In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

- (i) A relation R on the set of real numbers $\mathbb{R}$ is defined as $R = \{(a, b) : a \leq b^3\}$. Then R is: [1]
(Analyse)
- (a) reflexive only
(b) symmetric only
(c) transitive only
(d) neither reflexive, nor symmetric, nor transitive
- (ii) The principal value of $\sin^{-1}(\sin 3\pi/5)$ is: (Understand) [1]
- (a) $3\pi/5$
(b) $2\pi/5$
(c) $-2\pi/5$
(d) $\pi/10$

- (iii) If A is a square matrix of order 3 and $|A| = -2$, then $|3 \operatorname{adj} A|$ is equal to: (*Apply*) [1]
- (a) 108
 - (b) -108
 - (c) -54
 - (d) 36
- (iv) The radius of a circular metal plate is increasing at the rate of 0.2 cm/s when it is heated. The rate at which its area is increasing when the radius is 5 cm is: (*Apply*) [1]
- (a) $\pi \text{ cm}^2/\text{s}$
 - (b) $2\pi \text{ cm}^2/\text{s}$
 - (c) $10\pi \text{ cm}^2/\text{s}$
 - (d) $0.4\pi \text{ cm}^2/\text{s}$
- (v) Let $f(x) = x^3 - 3x^2 + 3x$, $x \in \mathbb{R}$. [1]
- Statement I: f is strictly increasing on $\mathbb{R}$.
- Statement II: Since $f'(1) = 0$, f has a local extremum at $x = 1$.
- Which of the following is correct? (*Understand*)
- (a) Statement I is true and Statement II is false.
 - (b) Statement I is false and Statement II is true.
 - (c) Both the statements are true.
 - (d) Both the statements are false.
- (vi) The acute angle between the curves $y = x^2$ and $x = y^2$ at their point of intersection $(1, 1)$ is: [1]
- (*Apply*)
- (a) $\pi/2$
 - (b) $\tan^{-1}(3/4)$
 - (c) $\tan^{-1}(4/3)$
 - (d) $\pi/4$
- (vii) For $f(x) = 2x^3 - 9x^2 + 12x + 5$, which of the following is correct? (*Apply*) [1]
- (a) f has a local maximum at $x = 1$ and a local minimum at $x = 2$.
 - (b) f has a local minimum at $x = 1$ and a local maximum at $x = 2$.
 - (c) f has local maxima at both $x = 1$ and $x = 2$.
 - (d) f has no local extremum, since $f''(x) = 0$ at $x = 3/2$.

(viii) The expression for the area of the shaded region shown below is: (*Apply*)

[1]



(a) $\int_0^2 x^2 dx$

(b) $\int_0^4 \sqrt{y} dy$

(c) $\int_0^2 (4 - x^2) dx$

(d) $\int_0^4 (\sqrt{y} - 2) dy$

(ix) Shown below is a solution of $\int x e^x dx$:

[1]

Step 1 $\int x e^x dx = x \int e^x dx - \int \left[\frac{d}{dx}(x) \cdot \int e^x dx \right] dx$

Step 2 $= x e^x - \int e^x dx$

Step 3 $= x e^x + e^x + C$

In which step is there an error (if any)? (*Understand*)

(a) Step 1

(b) Step 2

(c) Step 3

(d) No error

(x) On evaluating $\int \sin x \cos x dx$, Riya's answer was $\frac{1}{2} \sin^2 x + c$ and Kabir's answer was $-\frac{1}{2} \cos^2 x + c$. Whose answer is correct? (*Understand*)

[1]

(a) Only Riya's answer is correct.

(b) Only Kabir's answer is correct.

(c) Both of them are correct.

(d) Neither of them is correct.

(xi) If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 6$, then $|\vec{a} \times \vec{b}|$ is equal to: (*Apply*)

[1]

(a) 6

(b) $6\sqrt{3}$

(c) 12

(d) $3\sqrt{3}$

- (xii) If a line makes angles α, β, γ with the positive directions of the coordinate axes, then $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is equal to: (*Understand*) [1]
- (a) 1
(b) -1
(c) 2
(d) -2
- (xiii) The corner points of the feasible region of an LPP are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). The minimum of $Z = 4x + 6y$ occurs at: (*Analyse*) [1]
- (a) (0, 2) only
(b) (3, 0) only
(c) the mid-point of the line segment joining (0, 2) and (3, 0) only
(d) every point of the line segment joining (0, 2) and (3, 0)
- (xiv) Statement I: The feasible region of every linear programming problem is bounded.
Statement II: If the feasible region of an LPP is bounded, the objective function $Z = ax + by$ has both a maximum and a minimum value on it.
Which of the statements is correct? (*Understand*) [1]
- (a) Statement I is true and Statement II is false.
(b) Statement I is false and Statement II is true.
(c) Both the statements are true.
(d) Both the statements are false.
- (xv) If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$, then $P(A' | B)$ is equal to: (*Apply*) [1]
- (a) 0.4
(b) 0.6
(c) 0.5
(d) 0.3
- (xvi) Assertion: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3 + 1$ is a bijection.
Reason: Every strictly increasing function from $\mathbb{R}$ to $\mathbb{R}$ is onto.
Which one of the following is correct? (*Analyse*) [1]
- (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(c) Assertion is true and Reason is false.
(d) Assertion is false and Reason is true.
- (xvii) Assertion: $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \pi/4$.
Reason: If f is continuous on $[0, a]$, then $\int_0^a f(x) dx = \int_0^a f(a - x) dx$.
Which one of the following is correct? (*Understand*) [1]
- (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
(c) Assertion is true and Reason is false.
(d) Assertion is false and Reason is true.

- (xviii) For real numbers, $ab = 0$ implies $a = 0$ or $b = 0$. Give an example of two non-zero matrices A and B of order 2 such that $AB = O$, to show that this is not true for matrices. (*Understand*) [1]
- (xix) Find the sum of the order and the degree of the differential equation $[1 + (\frac{dy}{dx})^2]^{3/2} = 5 \frac{d^2y}{dx^2}$. [1]
(*Apply*)
- (xx) A random variable X takes the values 0, 1, 2, 3 with $P(X = x) = k(x + 1)$. Find the mean $E(X)$. [1]
(*Apply*)

SECTION B – 14 MARKS

Question 2

Find the angle between the line $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and the plane $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 4$. [2]
(*Apply*)

Question 3

On each of 6 consecutive days a street vendor either gets a bulk order or does not, the two being equally likely and the days independent. Find the probability that he gets bulk orders on at least 4 consecutive days. (*Evaluate*) [2]

Question 4

Solve the differential equation $(x^2 + y^2) dx - 2xy dy = 0$. (*Apply*) [2]

Question 5

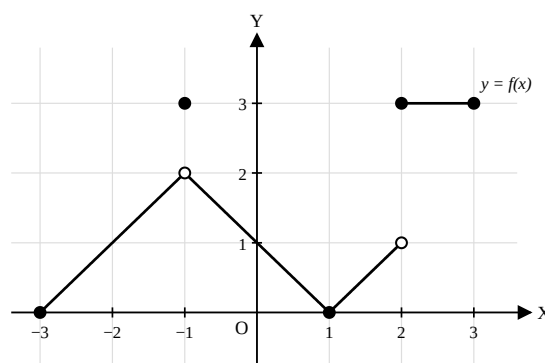
Determine the values of the constants p and q such that the function [2]

$$f(x) = \begin{cases} p e^x + q, & x \leq 0 \\ 2x^2 + 3x + 4, & x > 0 \end{cases}$$

is differentiable at $x = 0$. (*Understand*)

OR

Observe the graph of $y = f(x)$, $-3 \leq x \leq 3$, given below.



- (a) State the value(s) of x where f has a removable discontinuity. (*Understand*) [1]
- (b) State the value(s) of x where f is continuous but not differentiable. (*Understand*) [1]

Question 6

Solve for x : $\tan^{-1}(2x) + \tan^{-1}(3x) = \pi/4$. (*Apply*) [2]

OR

Prove that $2\tan^{-1}(1/2) + \tan^{-1}(1/7) = \tan^{-1}(31/17)$. (*Apply*) [2]

Question 7

Find the scalar projection and the vector projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. [2]
(*Apply*)

OR

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$, find a unit vector perpendicular to both $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$. [2]
(*Apply*)

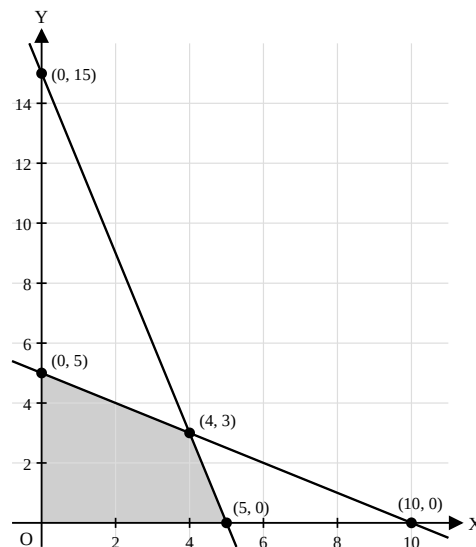
Question 8

A town map uses the market O as origin, with $\hat{i}$ a 1 km vector pointing east and $\hat{j}$ a 1 km vector pointing north. The railway station A is 6 km east and 2 km north of O, and the school B is 2 km east and 4 km north of O. A bus stop S is at the mid-point of the straight road OA.

- (i) Find the position vector of S and the vector $\vec{BS}$. (*Understand*) [1]
(ii) Show that S is the point on road OA nearest to the school, and find that distance. (*Understand*) [1]

SECTION C – 21 MARKS**Question 9**

The feasible region of an LPP is the shaded region shown below.



- (i) Write the constraints that determine the feasible region. (*Understand*) [1]
(ii) Hence maximise $Z = 3x + 2y$. (*Apply*) [1]
(iii) If the objective function is $Z = px + qy$ ($p, q > 0$) and its maximum occurs at both (5, 0) and (4, 3), find the relation between p and q . (*Analyse*) [1]

Question 10

If $y = (\sin^{-1}x)^2$, prove that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$. (Apply) [3]

Question 11

Let $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$, $x \neq 0$.

- (i) Find $\frac{dy}{dx}$. (Evaluate) [2]
- (ii) A student claims that y has a local maximum at some point $x = c > 0$. Using (i), evaluate the claim. (Evaluate) [1]

Question 12

Find the equation of the plane passing through the point $(-1, 3, 2)$ and perpendicular to each of the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$. Write it in vector form also. (Apply) [3]

OR

A line L passes through $A(3, -4, -5)$ and $B(2, -3, 1)$.

- (a) Write the Cartesian equation of L . (Understand) [1]
- (b) Find the point P where L meets the plane $2x + y + z = 7$. (Apply) [1]
- (c) Find the distance of P from the point $C(3, 4, 4)$. (Analyse) [1]

Question 13

Using properties of determinants, prove that [3]

$$\begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3.$$

(Analyse)

OR

Three stationery shops P, Q and R sold notebooks, pens and school bags in a week as follows: [3]
 P sold 30 notebooks, 40 pens and 10 bags; Q sold 20 notebooks, 50 pens and 20 bags; R sold 40 notebooks, 30 pens and 15 bags. A notebook costs ₹40, a pen ₹10 and a bag ₹250. Using a single matrix product, find the sales revenue of each shop and the total revenue. Which shop earned the most? (Apply)

Question 14

Let f be a continuous function such that $\int_{-2}^4 f(x) dx = 9$ and $\int_1^4 f(x) dx = 5$.

- (i) Find $\int_{-2}^1 f(x) dx$. (Understand) [1]
- (ii) Hence evaluate $\int_{-2}^1 (2f(x) - x) dx$. (Apply) [2]

Question 15

Using integration, find the area of the region bounded by the parabola $y^2 = 4x$ and the line $y = 2x$. (*Apply*) [3]

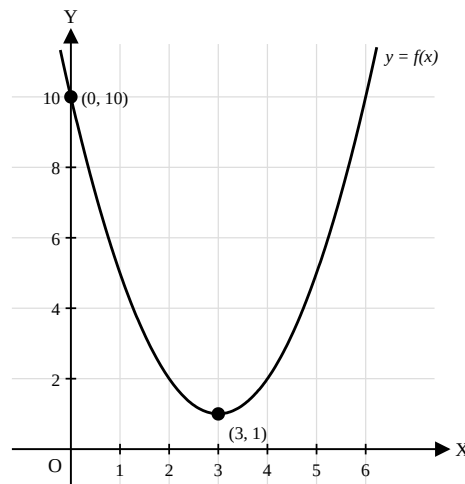
OR

Consider the curve $y = |x - 1|$.

- (a) Sketch the region bounded by $y = |x - 1|$, the x-axis and the lines $x = -2$ and $x = 3$. (*Understand*) [1]
- (b) Hence, using integration, find the area of the region. (*Apply*) [2]

SECTION D – 25 MARKS**Question 16**

The profile of a stretch of cycle track is modelled by $f(x) = x^2 - 6x + 10$, $x \in \mathbb{R}$, whose graph is shown below.



- (i) Find the range of f . (*Understand*) [1]
- (ii) Is $f : \mathbb{R} \rightarrow [1, \infty)$ invertible? Justify. (*Analyse*) [1]
- (iii) Let g be f restricted to $[3, \infty)$, $g : [3, \infty) \rightarrow [1, \infty)$. Find $g^{-1}(x)$. (*Apply*) [1]
- (iv) Let h be f restricted to $(-\infty, 3]$. Find $h^{-1}(x)$ and verify that $(h \circ h^{-1})(5) = 5$. (*Apply*) [1]
- (v) Find the value of x for which $g^{-1}(x) = h^{-1}(x)$. (*Analyse*) [1]

Question 17

The height (in hundreds of metres) of a hill road above a reference level is $h(x) = x^3 - 6x^2 + 9x + 2$, where x (in km) is the horizontal distance from the start, $0 \leq x \leq 4$.

- (a) Find the intervals on which the road is climbing and descending. (*Understand*) [1]
- (b) Find the points on the road where it is momentarily level. (*Understand*) [1]
- (c) Find the absolute maximum and absolute minimum heights of the road on $[0, 4]$. (*Analyse*) [2]
- (d) Find the equation of the tangent to the curve $y = h(x)$ at $x = 2$. (*Apply*) [1]

OR

A right circular cone is inscribed in a sphere of radius 9 cm.

- (a) If h is the height of the cone, express its volume V as a function of h . (*Apply*) [2]

- (b) Find dV/dh . (*Understand*) [1]
- (c) Find the height for which the volume is maximum, and the maximum volume. (*Evaluate*) [2]

Question 18

- (a) Evaluate $\int \frac{1}{x^2 + 4} dx$. (*Understand*) [2]
- (b) Hence evaluate $\int \frac{e^x}{e^{2x} + 4e^x + 8} dx$. (*Evaluate*) [3]

OR

- (a) Evaluate $\int x e^x dx$. (*Understand*) [2]
- (b) Hence evaluate $\int x^3 e^{x^2} dx$. (*Evaluate*) [3]

Question 19

In a factory, machines A, B and C produce 25%, 35% and 40% of the total output of bolts. Of their outputs 5%, 4% and 2% respectively are defective. A bolt is drawn at random from the day's production.

- (a) Find the probability that the bolt is defective. (*Apply*) [2]
- (b) The bolt drawn is found to be defective. Find the probability that it was made by machine B. (*Analyse*) [2]
- (c) Which machine is most likely to have produced this defective bolt? (*Evaluate*) [1]

OR

A blood test is 99% effective in detecting a certain disease when it is present, but it also gives a false positive result for 0.5% of healthy persons tested. 0.1% of the population actually has the disease. A person is tested at random.

- (a) Find the probability that the test is positive. (*Apply*) [1]
- (b) Given that the test is positive, find the probability that the person has the disease. (*Analyse*) [2]
- (c) Given that the test is negative, find the probability that the person does not have the disease. (*Evaluate*) [2]

Question 20

Mrs Sen invested a total of ₹50,000 in three schemes paying simple interest at 6%, 8% and 9% per annum. Her total annual interest is ₹3,850. The combined amount in the 6% and 9% schemes exceeds the amount in the 8% scheme by ₹10,000. Taking the amounts (in ₹ thousand) as x , y and z , form the system of equations, write it as $AX = B$, and solve it using A^{-1} to find the amount invested in each scheme. (*Evaluate*) [5]

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